

ABM BUSINESS LOGIC BIBLE

§16 — TIMING MATRIX (Every Clock in the System)

V2 Section 16 · One Master Reference · Consolidated From All Fifteen Prior Sections

Every section built since §3 introduced its own timers — polling cadences, freshness windows, SLAs, cooldowns, residence-time limits, review cycles. None of them were ever assembled into one place. §16 is that single reference: every clock in the system, organized by what it governs, with its originating rule cited so this table is never the authoritative source — the originating section always is — but is the fastest way to answer 'how long does X take' without searching fifteen documents.

Timers consolidated	60+	Source sections	15 (§1–§15)
New rules	6	Conflicts found	2 flagged

WHY THIS IS A CONSOLIDATION, NOT A NEW SPEC

Almost nothing below is new business logic — every timer cites the section that actually defines it. §16's job is purely to make fifteen documents' worth of clocks queryable in one table, and — in the process of consolidating — to catch the handful of places where two sections specify overlapping timers that don't quite agree. That discovery process is where this document earns its 6 new rules.

16.0 SIGNAL ACQUISITION & POLLING CADENCE

Timer	Duration	Source	What happens at expiry
Signal polling — Google News	Every 4 hours	V2 §3.1 / §4 v2 INT-SIG	New poll cycle starts
Signal polling — SAMA RSS	Every 6 hours	V2 §3.1	New poll cycle starts
Signal polling — LinkedIn	Every 12 hours	V2 §3.1	New poll cycle starts
Signal polling — Funding DBs	Every 24 hours	V2 §3.1	New poll cycle starts
INT-SIG SIG-NEWS/EVENT (deep)	Hourly / Daily	§4 v2 INT-SIG acquisition	Refresh acquisition cycle
INT-SIG SIG-EXEC/VENDOR/SUBS	Weekly + event-driven / Monthly + event-driven	§4 v2 INT-SIG acquisition	Refresh + event-triggered update
SIG-PATH (connection resolution)	Continuous	§4 v2 INT-SIG acquisition	Re-resolved on every relevant event
CRM deal-stage / meeting-outcome pull	Every 30 minutes	V2 §12.2 / §12 v1 CRM-PULL-01	Pull cycle re-runs
CRM manual-contact-addition pull	Every 60 minutes	V2 §12.2	Pull cycle re-runs

16.1 SIGNAL FRESHNESS LIFECYCLE

Status	Age window	Scoring weight	Source
FRESH	0–7 days	100%	V2 §3.6 / §4 v1 SCORE-SIG-02
AGING	8–21 days	70%	V2 §3.6
STALE	22–45 days	30%	V2 §3.6
EXPIRED	46+ days	0% (excluded from scoring)	V2 §3.6

Signal type	Dedup freshness window	Source
LEADERSHIP_HIRE	30 days	V2 §3.5 SIG-DEDUP
DIGITAL_INITIATIVE / API_MODERNIZATION / PARTNERSHIP	14 days	V2 §3.5 SIG-DEDUP
REGULATORY / OPEN_BANKING	7 days	V2 §3.5 SIG-DEDUP
FUNDING / LICENSING	30 days	V2 §3.5 SIG-DEDUP
HIRING_SURGE / PRODUCT_LAUNCH	14 days	V2 §3.5 SIG-DEDUP

SEMANTIC HALF-LIFE (§6 v3 LAYER 5) — GENERALIZES THE ABOVE

The freshness lifecycle above is the V2-era, single-curve model. §6 v3 ENR-DECAY-01 generalizes it into four category-specific decay curves — Operational (7–30d), Tactical (30–90d), Strategic (6–12mo), Structural (3–5y) — because a contact email and an organization's core market identity should not decay on the same clock. §6 v2 ENR-REF-DEEP-04 already turned the 30-day HOT refresh into a FLOOR rather than a fixed cadence for exactly this reason (see §16.2 below for the conflict this created).

16.2 ENRICHMENT & REFRESH CADENCE

Timer	Duration	Source	Note
Enrichment API timeout	30 seconds, retry once after 60s	V2 §6.2 Step 1	—
Email verification timeout	15 seconds, retry next cycle	V2 §6.2 Step 3	—
HOT full enrichment refresh	Every 30 days	V2 ENR-REF-001	Superseded as FLOOR, see below
WARM light enrichment refresh	Every 90 days	V2 ENR-REF-002	Company data only, no contact re-discovery
COLD enrichment	None	V2 ENR-REF-003	—
Verified-email TTL	30 days	§6 v1 ENR-EV-06	Re-verify before reuse past this window
Event-driven re-enrichment (exec move, vendor event, etc.)	Immediate, scoped to affected section	§6 v2 ENR-REF-DEEP-01	Partial refresh, not full cycle
Ramadan/Eid decay-rate adjustment	1.5× normal decay (Operational/Tactical only)	§6 v2 ENR-REF-DEEP-03	Does not apply to Structural facts
Enrichment stopping condition (HOT default)	≥80% committee coverage + ≥2 reachable + ≥1 path	§6 v3 ENR-STOP-01	Enrichment halts when met

CONFLICT FLAGGED — §16-CONFLICT-01: 30-DAY CADENCE vs FLOOR

§6 v2's own OPEN-Q6.4 already identified this: V2's literal text treats 30 days as a fixed CADENCE (re-enrich every 30 days regardless); §6 v2 ENR-REF-DEEP-04 turns it into a FLOOR (never exceed 30 days without refresh, but event-driven triggers may refresh sooner AND may make the fixed 30-day re-run unnecessary if a scoped refresh already happened). This table follows §6 v2's FLOOR interpretation as current, consistent with OPEN-Q6.4's still-open phased-rollout default — NOT a new resolution, just citing which interpretation is operative pending that question's resolution.

16.3 OUTREACH SEQUENCE TIMING

Touch/Step	Timing	Send window (KSA)	Source
Email Touch 1	Day 1 (immediate after approval)	Tue–Thu, 9–11am	V2 \$9.2 / \$9 v1 OUT-SEQ-EMAIL
Email Touch 2	Day 4 (+3d)	Sun–Thu, 9am–2pm	V2 \$9.2
Email Touch 3	Day 8 (+4d)	Sun–Thu, 9am–2pm	V2 \$9.2
Email Touch 4	Day 14 (+6d)	Tue–Thu, 9–11am	V2 \$9.2
LinkedIn Step 1 (connection)	Day 1, or Day 4 if dual-channel	—	V2 \$9.3
LinkedIn Step 2 (value DM)	+2 days after accept	—	V2 \$9.3
LinkedIn Step 3 (follow-up)	+7 days after Step 2	—	V2 \$9.3
Cross-channel gap (email→LinkedIn)	72 hours	V2 OUT-CHAN-002	—
LinkedIn connection ignored timeout	7 days → mark 'connection_pending'	V2 OUT-SEQ-005	Retry only on new P1 signal, min 30-day gap
Sequence-to-sequence cooldown (same contact)	45 days minimum	V2 OUT-SEQ-006	—
Cross-account same-person touch gap	14 days minimum	V2 EDGE-SYS-003	Same person, two accounts
Max outreach velocity (MVP, system-wide)	15 emails/day	V2 OUT-SEQ-007	Shared budget; allocation via \$5 v3 TIER-PORT-002 ranking (\$9 v2 OUT-PORT-02)

16.4 DELIVERABILITY & DOMAIN WARMUP

Timer	Schedule	Source
Domain warmup Week 1	Max 5 emails/day	V2 OUT-DELIV-002
Domain warmup Week 2	Max 10 emails/day	V2 OUT-DELIV-002
Domain warmup Week 3	Max 15 emails/day	V2 OUT-DELIV-002
Domain warmup, stabilized	Max 20 emails/day	V2 OUT-DELIV-002
Bounce-rate monitoring window	7 days; >5% → pause all sending from domain	V2 OUT-DELIV-004

16.5 ENGAGEMENT, QC & HUMAN REVIEW SLAs

Timer	Duration	Source
Sentiment AI confidence threshold escalation	Below 0.7 → human classification	V2 §10.2 ENG-SENT-05
AI QC regeneration attempts (Agent 6 level)	Max 3, then MANUAL_CREATION	V2 QC-REGEN-001 / §8 v1 QC-REGEN-01
AI QC regeneration (Agent 5 / strategy level)	Max 2, then MANUAL_CREATION	§8 v1 QC-REGEN-04
Human review SLA (business hours)	4 hours, Sun–Thu 9am–6pm KSA	V2 QC-HUMAN-003 / §8 v1 QC-HUMAN-03
Human review SLA (outside business hours)	Reviewed at start of next business day	V2 QC-HUMAN-003
Post-MVP human-review relaxation eligibility	After 90 days, ≥85% AI pass + <10% override, per segment×language	V2 QC-HUMAN-004 / §8 v1 QC-HUMAN-05
Handoff response SLA — positive reply / LinkedIn reply (URGENT)	2 hours, business hours	V2 §11.1 HAND-TRIG
Handoff response SLA — meeting request (CRITICAL)	1 hour	V2 §11.1
Handoff response SLA — pricing question (HIGH)	4 hours	V2 §11.1
Handoff response SLA — security/compliance (HIGH)	24 hours	V2 §11.1
Handoff response SLA — referral (MEDIUM)	New outreach within 24 hours	V2 §11.1
Handoff acknowledgment escalation	48 hours → escalate to sales manager	V2 HAND-006 / §11 v1 HAND-WF-06
Blind review audit cadence	Periodic (per §8 v2 QC-HUMAN-10)	§8 v2

16.6 TIER, ACCOUNT & OPPORTUNITY STATE TIMERS

Timer	Duration	Source
WARM 48h enrichment check (light)	48 hours after COLD→WARM	V2 TIER-UP-001
Manual tier-override expiry	30 days, then scoring engine resumes authority	V2 TIER-OVR-001 / §5 v1 TIER-OVR-02
Tier-change cooldown (OPEN-Q9.13 / OPEN-Q13 safe default)	24 hours after a promotion	§5 v2 OPEN-Q13 safe default
Engagement-free WARM→COLD demotion window	30 days with no engagement events	V2 TIER-DOWN-002
Dormant cooldown (standard)	90 days + new P1 signal required	V2 §14.1 / §12 v1 CRM-PULL-04
Dormant cooldown (competitor-won)	180 days (double standard)	V2 EDGE-SIG-003
HOT_WAITLIST timeout (OPEN-Q14.1 safe default)	Escalate to GOV review, not auto-demote, if persistently waitlisted	§14 v1 OPEN-Q14.1
RELATIONSHIP_MONITOR re-engagement check	New signal (immediate) + periodic backstop review	§14 v1 OPEN-Q14.2
ENRICHING state residence (expected)	≤ 48 hours	§14 v2 STATE-TIME-01
OUTREACH_ACTIVE state residence (expected)	≤ 30 days	§14 v2 STATE-TIME-01
HOT_UNENRICHED state residence (expected)	≤ 14 days	§14 v2 STATE-TIME-01
Opportunity OPP_IDENTIFIED residence	≤ 14 days	§14 v2 (Part B state table)
Opportunity OPP_QUALIFYING residence	≤ 30 days	§14 v2
Opportunity OPP_ACTIVE residence	≤ 60 days	§14 v2
Opportunity OPP_EVALUATING residence	≤ 90 days	§14 v2
Opportunity OPP_PROPOSAL residence	≤ 45 days	§14 v2
Opportunity OPP_NEGOTIATION residence	≤ 60 days	§14 v2
Opportunity PAUSED review cadence	Every 30 days	§14 v2 (Part B state table)

16.7 ANALYTICS, LEARNING LOOPS & GOVERNANCE CYCLES

Timer	Cadence	Source
Loop 1 — Signal Quality review	Every 30 days	V2 §13.2 / §13 v1 ANL-LOOP1-01
Loop 2 — Prompt Optimization review	Every 30 days	V2 §13.2 / §13 v1 ANL-LOOP2-01
Loop 3 — Scoring Weights recalibration	Every 90 days	V2 SCORE-RECAL-001 / §13 v1 ANL-LOOP3-01
Loop 4 — Persona Performance review	Every 30 days	V2 §13.2 / §13 v1 ANL-LOOP4-01
Loop 5 — Trust & Reputation Calibration (new)	Every 90 days	§13 v1 ANL-LOOP5-01
Weight-change max adjustment per cycle	±5% per dimension	V2 SCORE-RECAL-001
KPI Gaming Investigation	Quarterly	§13 v2 ANL-GUARD-04
Learning Review Board (cumulative-drift review)	Quarterly	§13 v2 ANL-LCHANGE-02
Resilience Investment quarterly review	Quarterly	§15 v2 FAIL-INVEST-03
Learning Horizon — Short KPIs	Days	§13 v2 ANL-HORIZON-01
Learning Horizon — Medium KPIs	Weeks to a quarter	§13 v2 ANL-HORIZON-01
Learning Horizon — Long KPIs (Reputation Yield, Convergence)	6–18 months	§13 v2 ANL-HORIZON-01
Reputation Yield quantification graduation (OPEN-Q13.3)	Pending sufficient Loop 5 longitudinal data	§13 v1 OPEN-Q13.3

16.8 RETRY, FAILURE & RECOVERY TIMERS

Failure type	Retry schedule	Max retries	Source
API timeout	30s, 1m, 5m, 15m	4	V2 §15.1 / §15 RETRY table
AI model failure	Retry same model after 60s, then fallback	2+1	V2 §15.1
Email delivery (soft bounce)	Retry once after 4h	1	V2 §15.1
LinkedIn API failure	Retry after 15 minutes	3	V2 §15.1
CRM sync failure	1m, 5m, 15m, 60m	4	V2 §15.1 / §4 v2 INT-CRM-03
Webhook (engagement) failure	Every 5 minutes for 1 hour	12	V2 §15.1
Dead-letter queue review	At least once per business day	—	V2 RETRY-002 / §12 v1 CRM-SYNC-03
Service failure-rate alert threshold	20% in any 1-hour window	—	V2 RETRY-003
Failure recurrence → mandatory Root Cause Review	3rd occurrence of same signature	—	§15 v2 FAIL-REPEAT-02

16.9 KSA BUSINESS-HOURS & CALENDAR CONSTRAINTS (PREVIEW — FULL TREATMENT IN §17)

PLACEHOLDER, NOT YET AUTHORITATIVE

The KSA-specific calendar constraints (Sunday–Thursday work week, 9am–6pm AST business hours, Ramadan volume/window reduction, Eid blackouts) are referenced throughout this matrix wherever a 'business hours' or 'KSA send window' timer appears, but their full specification — the exact Ramadan reduction percentage, the exact Eid blackout durations, the Vision 2030/SAMA CSF/PDPL constraints — belongs to §17, not here. This row exists so §16 doesn't silently omit the fact that every 'business hours' reference above is INCOMPLETE until §17 is built.

16.10 NEW RULES — WHAT CONSOLIDATION ITSELF DISCOVERED

Rule ID	Specification
TIME-MATRIX-01	This document (§16) is a CONSOLIDATED REFERENCE, not an authoritative source. Where this table and an originating section's text ever appear to disagree, the ORIGINATING SECTION governs; §16 is corrected to match, never the reverse.
TIME-MATRIX-02	Every timer in this matrix cites its originating rule ID and section; a timer with no citation is not yet permitted to exist in §16 — this prevents §16 from becoming a second, independently-evolving source of timing truth.
TIME-MATRIX-03	§16-CONFLICT-01 (30-day enrichment cadence vs floor, §16.2) is explicitly NOT resolved by this document — it inherits §6 v2 OPEN-Q6.4's still-open status. §16 surfaces the conflict for visibility; it does not adjudicate open questions that belong to their originating section.
TIME-MATRIX-04	A second minor inconsistency was found in cross-referencing: §5 v2's safe default for tier-change cooldown (24 hours, from OPEN-Q13 in that document) and §9 v2 Layer 9's Portfolio-based allocation (§16.3) both touch HOT-admission pacing but were specified independently. They are not contradictory — the cooldown gates HOW OFTEN a tier can change; the Portfolio ranking gates WHICH accounts get capacity when multiple qualify simultaneously — but §16 records this explicitly so the relationship is documented rather than assumed obvious.
TIME-MATRIX-05	§16 will require a maintenance pass whenever any future section introduces or changes a timer — this document is correct as of §1–§15's current state and is NOT self-updating; a process note, not a technical rule, but recorded here so the obligation isn't lost.
TIME-MATRIX-06	All KSA business-hours and calendar-dependent timers in this matrix (every 'business hours' and 'send window' reference) are flagged as PROVISIONAL pending §17's full specification of the underlying calendar constraints (§16.9) — §16 does not invent Ramadan/Eid adjustment factors; it defers them entirely to §17.

§16 COMPLETE — WHAT IT ESTABLISHED

§16 is now one queryable reference for every timer specified across fifteen prior sections — polling cadences, freshness windows, enrichment refresh, outreach sequencing, deliverability warmup, QC and human review SLAs, state-residence expectations, learning-loop cycles, and retry/recovery schedules. Nothing here is new business logic except the consolidation discipline itself: §16 cites its sources rather than restating them as independent truth, surfaces (without resolving) the one genuine timing conflict the consolidation process found (§6's 30-day cadence-vs-floor question, already an acknowledged open question in its origin section), and explicitly flags every KSA-calendar-dependent timer as provisional pending §17. 6 new rules, all about HOW this matrix relates to its sources, not new timing decisions of its own.

With §16 built, only §17 remains to bring the Bible to full V2 section-number coverage.

PLACEMENT MAP UPDATE

§16 status: Complete. §16-CONFLICT-01 is recorded as inheriting §6 v2 OPEN-Q6.4's open status, not a new open question of §16's own.

ON THE HORIZON

§17 KSA-Specific Business Rules — the Bible's last 'Not started' section, and per multiple independent assessments, likely the richest remaining territory: Sunday–Thursday work week, Ramadan/Eid blackouts, SAMA CSF, PDPL, Islamic banking terminology, Vision 2030 alignment. §17 will retroactively make every provisional KSA-calendar timer in this §16 matrix authoritative.